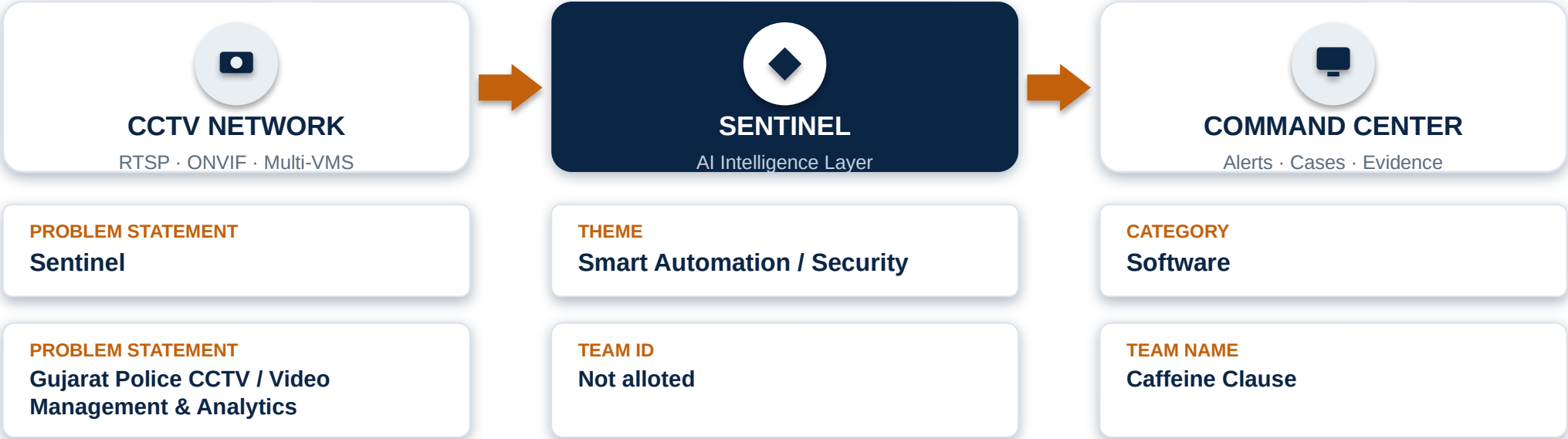


SENTINEL

AI-Powered Unified CCTV Intelligence & Investigation Platform

"80,000 Cameras. One Intelligence Layer."



TEAM MEMBERS

KJ Kavya Joshi Team Lead · Full-Stack	PR Prajinraj Ranawat AI · Computer Vision	VN Vanshal Nagar Backend · APIs	IM Isha Masani Frontend · UI/UX	VS Vishakha Sharma GIS · Data Engineering	RL Rishit Lad DevOps · Security
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FROM CCTV FOOTAGE TO ACTIONABLE INTELLIGENCE

Proposed Solution · built and demonstrated end-to-end

26+ DEPARTMENTS

Police · Traffic · Municipal · Transport

CCTV / VMS SOURCES

RTSP · HLS · ONVIF · Vendor VMS

SENTINEL INTELLIGENCE LAYER

Ingestion · Federation · Registry

AI DETECTION · ANPR · TRACKING

YOLO detection + OCR plates + ByteTrack

EVENT CORRELATION

Plate + time + camera + location

INVESTIGATION COPILOT & GRAPH

Natural-language search, timeline, entity graph

ALERTS → INCIDENTS → EVIDENCE → CASES

Case-ready, audited output

CORE INNOVATIONS

Unified CCTV Registry + GIS

Every camera mapped, searchable, health-monitored

Cross-Camera Vehicle Intelligence

Plate + appearance + time correlation across feeds

AI Investigation Copilot + Graph

Ask a question, get journey, map, timeline + graph

Real-Time Alerts & Evidence Chain

Watchlist hits to court-ready, audited evidence

WHY SENTINEL?

TODAY

- Camera-by-camera viewing
- Manual footage searching
- Isolated, unlinked alerts
- Fragmented evidence

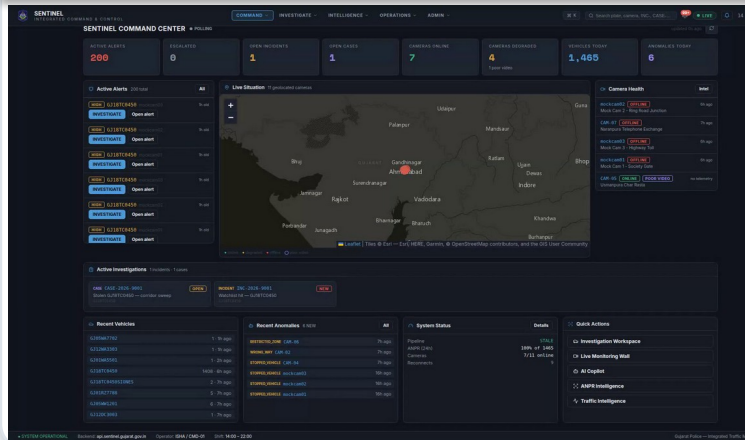
WITH SENTINEL

- Unified statewide search
- AI-assisted investigation + graph
- Correlated cross-camera events
- Case-ready evidence chain

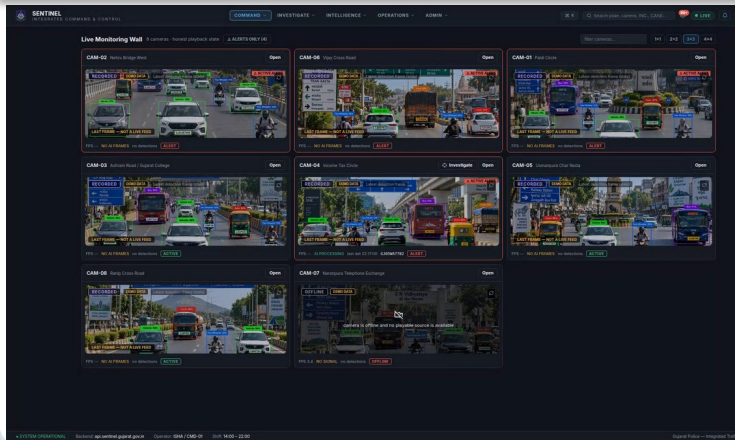
THE PLATFORM — LIVE, NOT MOCKED

Real screens from the running system — live seeded data, not static mockups

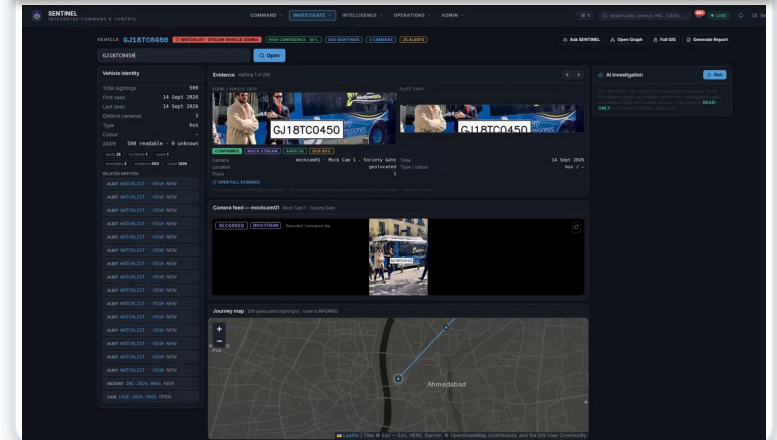
Command Center



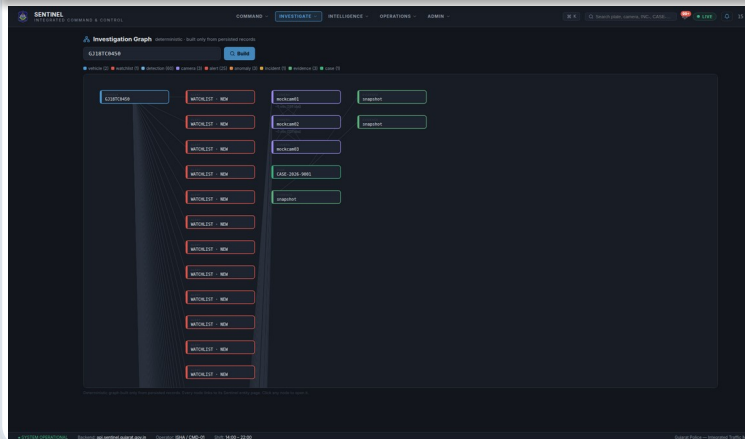
Live Monitoring Wall — real-time AI detection



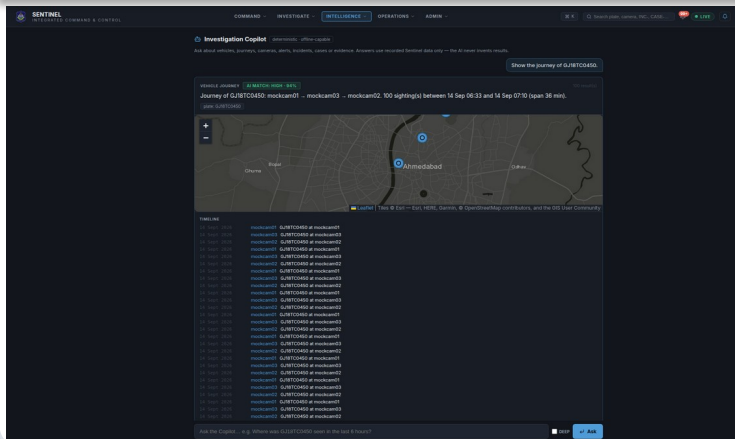
Vehicle Investigation Workspace



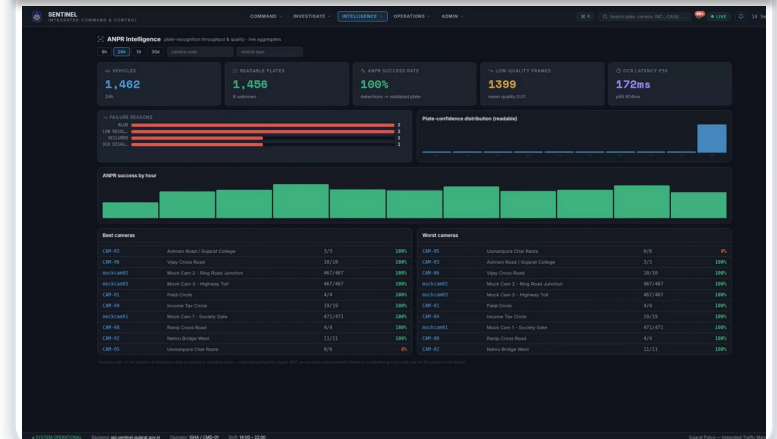
Investigation Graph



AI Investigation Copilot



ANPR Intelligence

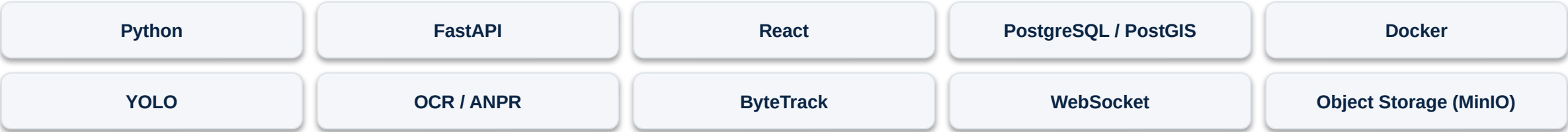


TECHNICAL APPROACH

End-to-end pipeline · technology stack · security by design



TECHNOLOGY STACK



SECURITY BY DESIGN

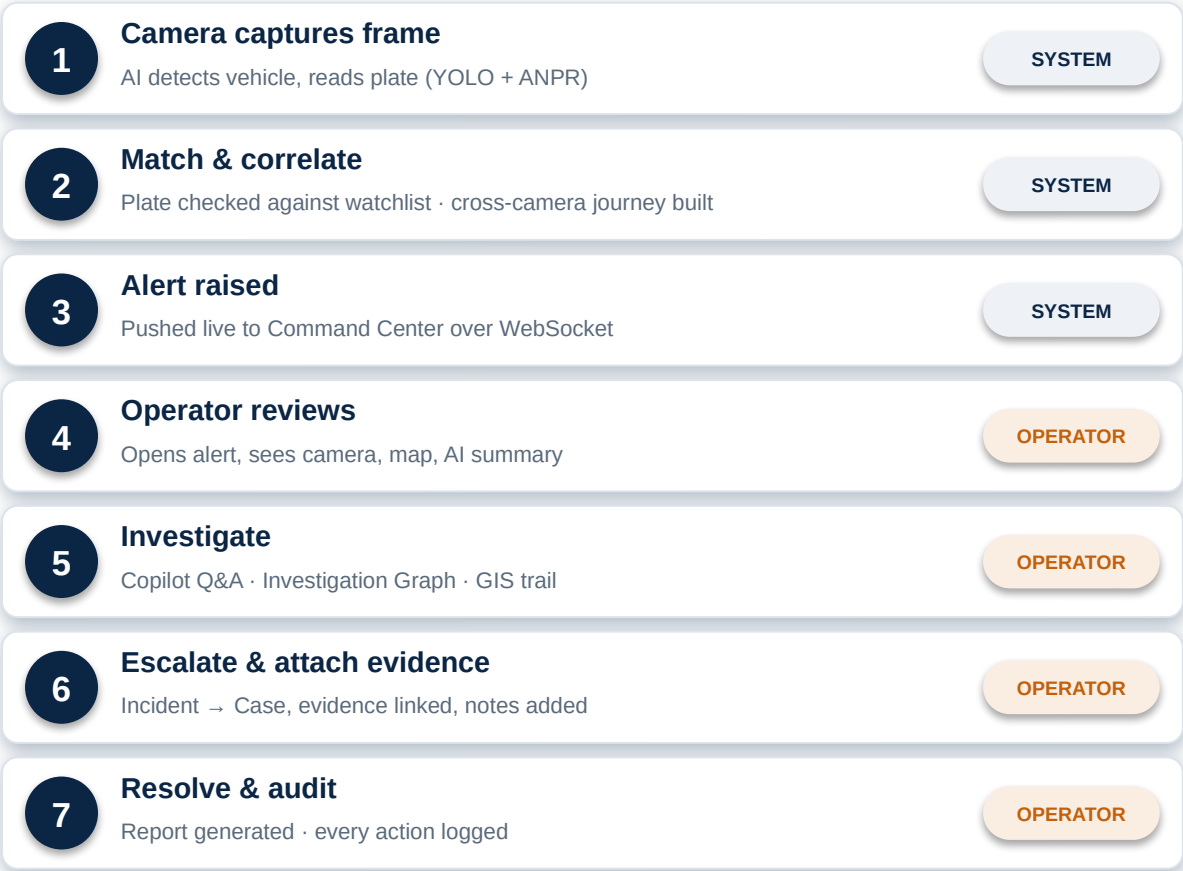


Metadata-first design — video stays at source, only events and evidence pointers move across the network.

WORKFLOW & SYSTEM INTEGRATION

How one detection becomes a resolved case — and what Sentinel connects to

END-TO-END OPERATIONAL WORKFLOW



SYSTEM INTEGRATION MAP



FEASIBILITY & VIABILITY

Buildable today · operable by police teams · designed to scale statewide

 TECHNICAL

- Vendor-neutral architecture
- Open standards: RTSP / ONVIF
- Modular AI microservices
- PostgreSQL / PostGIS core
- Edge + central processing ready

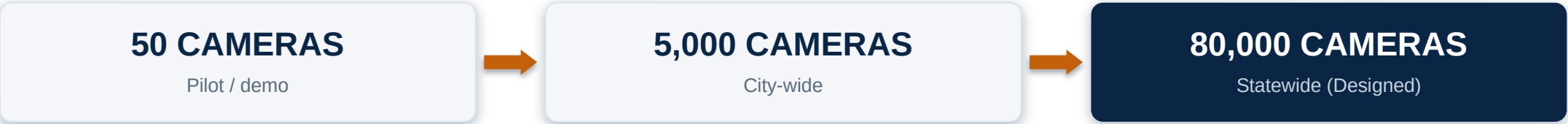
 OPERATIONAL

- Single command-centre interface
- GIS-based camera management
- Automated alerting workflows
- Investigation timelines & graph
- Evidence & report generation

 SCALABILITY

- Regional / edge AI processing
- Metadata-first architecture
- Load-balanced analytics workers
- Hot / Warm / Cold storage tiers
- Designed for 80,000-camera scale

SAME ARCHITECTURE — HORIZONTAL SCALE



Architecture layers stay constant — compute and storage scale out.

RISKS → MITIGATION

Bandwidth constraints
Edge processing at source

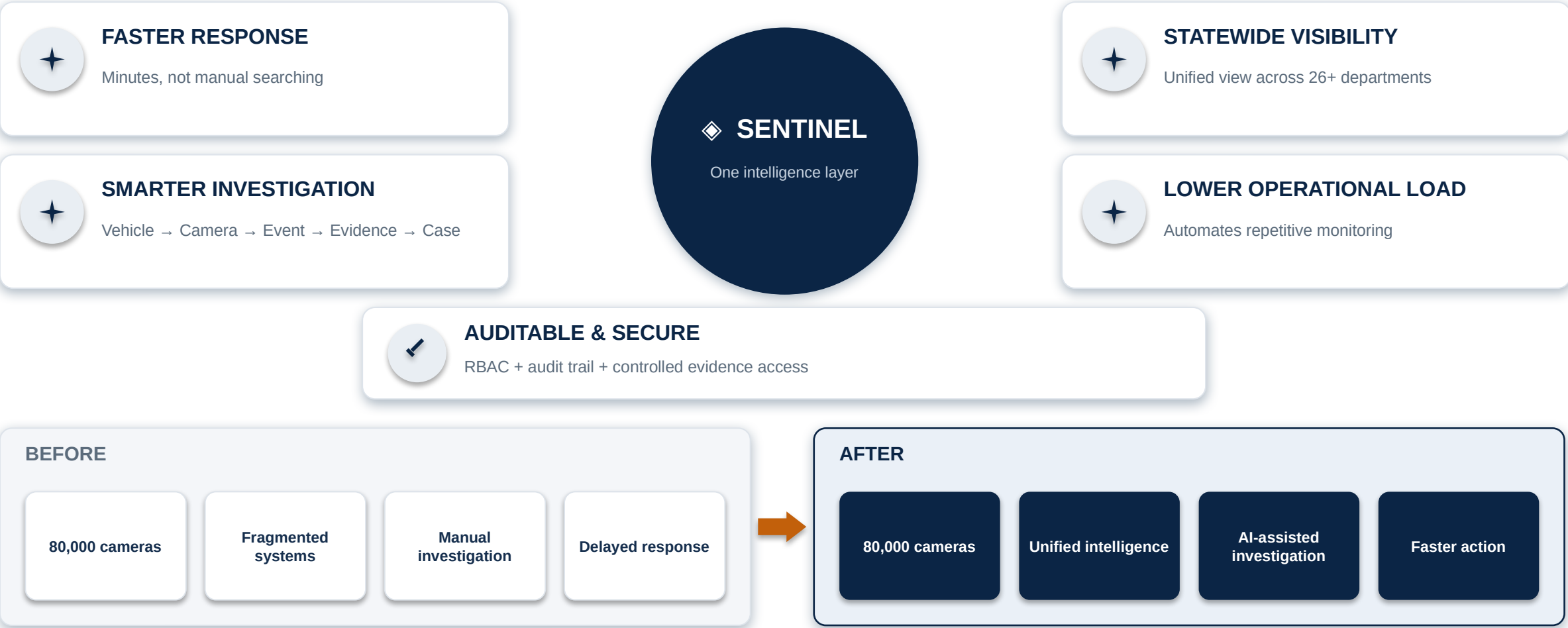
Vendor lock-in
Adapter / abstraction layer

AI false positives
Confidence score + human review

System failure
HA / DR architecture (Designed)

IMPACT & BENEFITS

From monitoring cameras to understanding events



"From monitoring cameras to understanding events."

RESEARCH & REFERENCES

Grounded in open standards, published CV research and proven distributed architecture

OFFICIAL / DOMAIN

- Gujarat Police CCTV / Sentinel ecosystem
- SIH 2026 Problem Statement SIH26174
- RTSP & ONVIF interoperability standards
- PostGIS geospatial architecture

AI / COMPUTER VISION

- YOLO real-time object detection
- ByteTrack multi-object tracking
- OCR / ANPR plate-recognition literature
- Cross-camera vehicle Re-ID research

ARCHITECTURE

- Event-driven video analytics patterns
- Edge + cloud video processing models
- High-availability distributed systems
- Secure RBAC & audit architectures

RESEARCH → SOLUTION

01



RESEARCH

Standards, CV papers,
police workflows

02



ARCHITECTURE

Vendor-neutral,
metadata-first design

03



PROTOTYPE

Working platform on
recorded + live feeds

04



DEPLOYMENT ROADMAP

Edge rollout, HA/DR,
dept. integrations